Personalized Clothing Recommendation System

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ABSTRACT

E-Commerce has seen a lot of growth over the past decade. With an increase in commodities, especially fashion accessories and clothing items in the online market, a need for an efficient recommender system arises. A number of algorithms exist for clothing recommendations in E-commerce. This research proposes a clothing recommendation system, which uses both image processing on product images and Natural Language Processing on user review to predict accurate user ratings. The model employs hybrid model which uses collaborative filtering and content based recommendation to predict user rating. To further alleviate the cold-start problem, knowledge graph is created from which hidden relationships can be extracted.

Background

An effective clothing recommender system would successfully catch on the user's shopping pattern and suggest products that are likely to interest the user, thus encouraging good shopping experience for the user and increasing gross merchandise value[1]. In order to build such a recommender system understanding both the dimensions of people's preferences as well as their dynamics is important[2]. User's personal aesthetic preferences also play a vital role, and these preferences may be highly influenced by their geographical locations and culture. Another key factor that must be taken into consideration is the user's budget and affordability.

Problem Addressed

The aim of this project is to explore recommender systems and various methods to build an effective personalized clothing recommendation system with better accuracy using images of products and customer reviews. Moreover, the model should be able to take care of the cold-start scenario since new items keep being frequently updated in the online market without any ratings or feedback.

Related Work

Most existing methods rely on conventional features to represent an image, such as the visual features extracted by convolutional neural networks (CNN features) and the scale-invariant feature transform algorithm (SIFT features), color histograms, and so on[345]. The related research shows that visual features based on deep learning can bring higher accuracy Recommendation Systems, and it is a trend to use visual features for fashion recommendation[67]. He et al presented novel models for the One-Class Collaborative Filtering setting, to estimate users' fashion-aware personalized ranking functions based on their past feedback, the method combines high-level visual features extracted from a deep convolutional neural network, users' past feedback, as well as evolving trends within the community[2].
Gu et al presented a fashion coordinate system that considers user behavior and visual fashion style. It extends latent factor model (LFM) to handle user behavior characteristics and uses the denoising autoencoder network model to process visual features[8]. Cheng et al proposed aspect-aware topic model (ATM) on the review text to model - user preferences and item features from various aspects and estimate the aspect importance of a user towards an item. The importance of the aspect is then integrated into a novel aspect-aware latent factor model (ALFM), which learns user's and item's latent factors based on ratings[9]. Wen et al in their work construct knowledge graph of user, knowledge graph of clothing and knowledge graph of context, utilize Apriori algorithm to capture the intrinsic correlations between clothing attributes and context attributes[10]. In the framework used by Yan et al, a data augmentation algorithm based on transfer learning is proposed to filter out the irrelevant items and label items with fine-grained tags, and a user-item knowledge graph is built to discover the potential relationship between items and users[1].

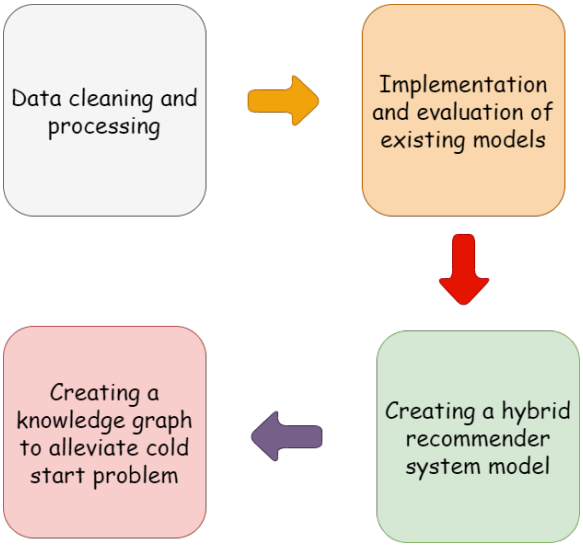


Figure 1. Design Approach

Method

The first step would be to study and analyze the Amazon Fashion dataset which would be used for this project, followed by data filtering and cleaning. Since there are already many recommendation models and techniques, they need to be implemented to evaluate their performance on the Amazon dataset. The model can be optimized after exploring various other recommendation models such as Aspect-Aware Latent Factor Model[9], a Deep Reinforcement learning model[11], Network-based recommendation model[12][13], Variational Autoencoders for Collaborative Filtering[14], etc.

This project proposes using a hybrid model which is based on both content-based recommender system and collaborative filtering. CNN architecture would be used to learn user's aesthetic preferences from their history. Natural language processing is to be used on user reviews, this data can further add value after converting it to vector using matrix factorization to capture important aspects from the textual data.

To deal with the cold start problem, the next step would be to create a knowledge graph from the amazon fashion dataset, this way we can apply graph algorithms to extract information, including entity extraction and relationship extraction. The design approach is depicted in Fig. 1.

Validation

Experimentally we evaluate our method dataset from Amazon.com. It's a classic e-commerce dataset, which includes 142.8 million reviews (ratings, text, helpfulness votes) spanning from May 1996 to July 2014, item metadata (descriptions, category information, price, brand, and image features). It contains high resolution image URLs for image processing[15]. Performance of the model will be evaluated based on how accurately the ratings are predicted.

References

1. Yan, C., Chen, Y. & Zhou, L. Differentiated fashion recommendation using knowledge graph and data augmentation. *Ieee Access* **7**, 102239–102248 (2019).
2. He, R. & McAuley, J. Ups and downs: Modeling the visual evolution of fashion trends with one-class collaborative filtering. In *proceedings of the 25th international conference on world wide web*, 507–517 (2016).
3. Yu, W. *et al.* Aesthetic-based clothing recommendation. In *Proceedings of the 2018 world wide web conference*, 649–658 (2018).
4. McAuley, J., Targett, C., Shi, Q. & Van Den Hengel, A. Image-based recommendations on styles and substitutes. In *Proceedings of the 38th international ACM SIGIR conference on research and development in information retrieval*, 43–52 (2015).
5. Wang, H., Zhou, Z., Xiao, C. & Zhang, L. Content based image search for clothing recommendations in e-commerce. In *Multimedia Data Mining and Analytics*, 253–267 (Springer, 2015).

6. Lin, Z. An empirical investigation of user and system recommendations in e-commerce. *Decision Support Systems* **68**, 111–124 (2014).

7. Bracher, C., Heinz, S. & Vollgraf, R. Fashion dna: merging content and sales data for recommendation and article mapping. *arXiv preprint arXiv:1609.02489* (2016).

8. Gu, S., Liu, X., Cai, L. & Shen, J. Fashion coordinates recommendation based on user behavior and visual clothing style. In *Proceedings of the 3rd International Conference on Communication and Information Processing*, 185–189 (2017).

9. Cheng, Z., Ding, Y., Zhu, L. & Kankanhalli, M. Aspect-aware latent factor model: Rating prediction with ratings and reviews. In *Proceedings of the 2018 world wide web conference*, 639–648 (2018).

10. Wen, Y., Liu, X. & Xu, B. Personalized clothing recommendation based on knowledge graph. In *2018 International Conference on Audio, Language and Image Processing (ICALIP)*, 1–5 (IEEE, 2018).

11. Zheng, G. *et al.* Drn: A deep reinforcement learning framework for news recommendation. In *Proceedings of the 2018 World Wide Web Conference*, 167–176 (2018).

12. Tang, J., Aggarwal, C. & Liu, H. Recommendations in signed social networks. In *Proceedings of the 25th International Conference on World Wide Web*, 31–40 (2016).

13. Su, J., Sharma, A. & Goel, S. The effect of recommendations on network structure. In *Proceedings of the 25th international conference on World Wide Web*, 1157–1167 (2016).

14. Liang, D., Krishnan, R. G., Hoffman, M. D. & Jebara, T. Variational autoencoders for collaborative filtering. In *Proceedings of the 2018 world wide web conference*, 689–698 (2018).

15. Ni, J., Li, J. & McAuley, J. Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 188–197 (2019).